Applying economics to the real world: model simulation vs econometrics

European Economics (ECON 2071)

The aim of these notes is to give you some introduction on the methods of analysis which are often used in the assigned journal articles, their purpose and the difference between them.

It is useful to start with a simple example. Let us think about the following practical economic question. Suppose that the government would be committed to spend 100 million pound (e.g. in the NHS) in the hope of increasing GDP. The government would like to know whether this is a viable policy and by how much GDP would increase. There are two possible ways to answer this question.

1 Model simulation

We need a theory that links government spending to GDP. Let us consider a theory that you already know, the Keynesian theory of aggregate demand. This states that aggregate demand Y is determined as follows:

$$Y = C + I + G + X - M \tag{1}$$

We assume that consumption C and imports M are a linear function of income:

$$C = cY \tag{2}$$

$$M = mY \tag{3}$$

c (m) is the propensity to consume (import). Investment (I), government expenditure (G) and exports (X) are all exogenous variables.

Substituting (2) and (3) into (1), we can show that aggregate demand is a function of the Keynesian multiplier $\left(\frac{1}{1-c+m}\right)$ and the autonomous components of demand G, I and X:

$$Y = \frac{1}{1-c+m}(G + I + X) \tag{4}$$

We have built a very simple theory on the relationship between aggregate demand and government expenditure, which says that an increase in government expenditure will increase aggregate demand (GDP):

$$\frac{\Delta Y}{\Delta G} = \frac{1}{1-c+m} \tag{5}$$

We are already able to tell the government that the idea behind the policy (increase spending to increase GDP) makes sense according to the model. However, the government would also like to quantify the effects of an increase in spending of 100 millions, as it aims for certain targets on growth etc. This is where model simulation becomes useful. In this simple model, it is straightforward to notice that this can only be done if we know the value of the Keynesian multiplier, which is a combination of two parameters of the model (c and m). But how do you know what is the value of the Keynesian multiplier in the UK? Economists either rely on other studies to set these values or they look at the data themselves. For instance, we might take data on GDP and consumption for our country for the last 20 years. Dividing consumption by GDP in each year, suppose that we observe that consumption tends to be a constant fraction of GDP and that fraction is 0.8. Suppose that we do the same for imports and we find that this value is 0.2. Therefore, we can set $c = 0.8$ and $m = 0.2$. This would give us a Keynesian multiplier of roughly 1.67. Therefore, if the government is committed to spending 100 million pound, we are able to say that GDP should increase by 167 millions from equation (5). Given equation (2), we are also able to say that aggregate consumption would increase by 133 millions.

The size of the Keynesian multiplier is still a very active research area in macroeconomics.

2 Econometrics

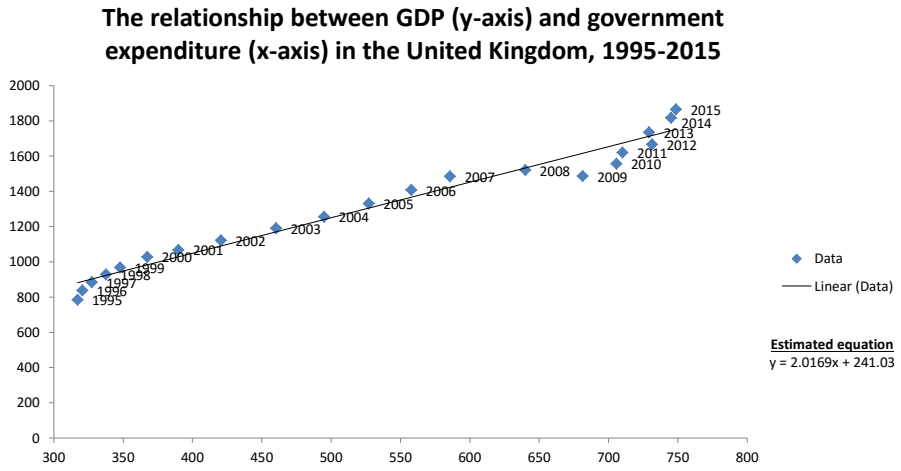
Another approach is to use the tool of econometrics, which is to look directly at the data. Our first problem is to select our sample, which involves selecting a country (countries) and a time period. For instance, we might want to test whether there is a relationship between GDP and government expenditure in the UK in the period between 1995 and 2015. Each point in the Figure shows how much was GDP and government expenditure in the UK for a particular year. The naked eye can already detect that there must be a positive relationship between the two variables. Usually, it is more complicated to detect whether there is any significan relationship. The approach is to specify a (usually) linear equation, where GDP is a function of government expenditure, as follows:

$$Y_i = a + bX_i + \varepsilon_i$$

where i stands for each data point and a is a constant. In this case, Y (endogenous variable) stands for GDP and X (exogenous variable) for government expenditure. ε is the error term, which simply recognize that there is going to be some randomness in the data, therefore the relationship will not

be perfect. The approach is then to "fit a line" or find the a and b which provides the best approximation of such relationship in our sample[1]. Notice that our approach does not restrict the sign of b to be positive. Although our guess is that b is going to be positive in this sample, we might get a negative b in another sample.

The software tells us that the estimated b (often called $\hat{b}$) is equal to 2.0169. Given our theoretical model, we may interpret this number as the Keynesian multiplier for the UK[2]. On the basis of our estimated equation (see Figure), we can tell the government that an increase of government expenditure of 100 million pound would increase GDP by 201.69 million pound. This is a higher number than the one that we found with model simulation. Before being confident in telling the government this result, there is also the important question as to whether the coefficient is actually statistically significant or not. In this case, it is. In other cases, our scatter plot looks more like a cloud of points: the software would tell us instead that $\hat{b}$ is not significant and the fitted line would be a straight line (X is not able to explain Y).



Source. World Economic Outlook Database (IMF). Gross domestic product, current prices (National currency) and General government total expenditure (National currency).

[1]This is a simple description of the main idea behind econometrics. For students who have never (and are not going to) come across a regression equation in their studies, I would suggest that you read the first chapters of any introductory econometrics textbook for more rigorous explanations. Jeffrey Wooldridge's Introductory Econometrics is available in the library and a very good textbook.

[2]Whenever we design an econometric model on the basis of a theoretical model, we usually say that it is a structural model.

3 Further comments

These are two complementary ways of bringing the theory to the data. Both can be used to make predictions/forecasting. Although both are extremely important and useful, I will now briefly discuss the advantages and disadvantages of each method.

It is easy to see that model simulation involves some kind of "faith" in the model used. However, an advantage of model simulation is that, in more complex models where the behaviour of key variables in the economy can be traced back to agents' decisions (micro-foundations), we are able to understand the mechanism by which a certain phenomenon occurs. Moreover, we are able to evaluate the welfare implications of certain policies through the impact on agents' utility. For this reason, model simulation is one of the most preferred tools for policy evaluation in international trade and macroeconomics.

Econometrics has instead the advantage of imposing less restrictions on the data. However, we need economic data in order to be able to run a regression model. For instance, Dhingra et al. (2016) have used model simulations to make predictions about the effects of Brexit. Once Brexit will have happened, we will be able to use the available data on trade, GDP, unemployment etc. to test with our econometric tools whether Brexit had any effect on these variables in the UK (and possibly on the EU as well).

Another issue with econometrics is whether the results obtained from our small sample can tell us something general about economic relationships or not. Economic models are more general in this sense. Finally, we should always be aware that there may be omitted variables in the econometric models which can bias our results.